Notes: Oehmke and Zawadowski (RFS, 2017)

The Anatomy of the CDS Market

Contents

1 Big picture

- **Question.** What motivates positions and trading in single-name corporate credit default swaps (CDSs), and what economic function do CDS markets actually perform?
- **Main idea.** CDS markets are not just insurance markets. They also act as *alternative trading venues* for credit risk when the underlying corporate bond market is fragmented, heterogeneous, and illiquid.
- **Core empirical challenge.** We want to know whether CDS activity reflects hedging, speculation, standardization/liquidity advantages, or arbitrage. These motives can all generate CDS positions, but they imply different cross-sectional patterns in CDS positions, CDS trading volume, and bond-market outcomes.
- **Key innovation in data.** The paper uses newly available disaggregated DTCC data on *net notional CDS positions* and *CDS trading volume* at the reference-entity level. That lets the authors move beyond aggregate BIS-style statistics and ask which firms attract more CDS activity and why.
- **Four economic motives tested.**
  - *Hedging:* firms with more insurable interest should have more CDS positions and more CDS trading.
  - *Speculation:* firms with more disagreement should attract more CDS activity.
  - *Standardization / liquidity:* CDSs should be used more when the underlying bonds are fragmented into many issues with heterogeneous terms.
  - *Arbitrage:* CDS positions should rise when the CDS–bond basis creates profitable basis trades.
- **Main result.** The strongest broad message is that CDS markets serve a *standardization and liquidity role*. Net CDS positions and CDS trading are larger when underlying bonds are split into many issues and differ in contractual features, and these same bonds are less liquid and more expensive to trade in the cash market.
- **Hedging result.** Firms with more outstanding bonds have larger net CDS positions. The baseline estimate implies about **7.77 cents** of net CDS per additional dollar of outstanding bonds.
- **Speculation result.** Analyst disagreement strongly predicts CDS activity, especially *CDS trading volume*. One standard deviation more disagreement raises monthly CDS trading by about **43.1%** for the median firm, but has little economically meaningful effect on bond trading.
- **Standardization result.** Above-median bond fragmentation is associated with about **$272 million** more net CDS positions, about a **33%** increase relative to the median firm. Contractual heterogeneity across bond issues is associated with about **$291 million** more net CDS positions, about a **36%** increase relative to the median firm.
- **Arbitrage result.** A more negative CDS–bond basis is associated with larger CDS positions. Basis traders lean against the negative basis, compress the mispricing, and reduce price impact in the bond market.

- **Policy/economic takeaway.** The paper paints CDS markets less as redundant side bets and more as a response to frictions in the cash bond market. CDSs standardize credit-risk trading, attract speculation to the more liquid venue, and help arbitrageurs absorb bond supply shocks.

2 Data and measurement (key objects)

2.1 Core datasets and sample construction

- **Primary CDS source.** The paper uses the DTCC Trade Information Warehouse (TIW), which captures about **95%** of globally traded CDSs.
- **Two main CDS objects.**
  - **Net notional CDS amount:** the stock of credit risk transferred via CDS positions.
  - **CDS trading volume:** the flow of credit risk transfer through trades that represent market-risk activity.
- **Other data sources.** Compustat for firm balance-sheet data and ratings; Mergent FISD for bond characteristics and amounts outstanding; TRACE Enhanced for bond trading volume and trading-cost measures; Markit for CDS quotes; CRSP for equity prices; and IBES for analyst forecasts.
- **Firm sample.** Public U.S. parent companies in Compustat that are traded reference entities in the CDS market, have at least one outstanding SEC-registered bond, are rated by S&P, and have enough analyst coverage to compute disagreement.
- **When a firm is treated as having a CDS market.** A firm is considered a traded CDS reference entity if it appears in DTCC or has at least three dealer CDS quotes in Markit. Once a CDS market exists for a firm, the paper assumes it continues to exist for the rest of the sample.
- **Sample period.**
  - Net notional CDS positions: **October 2008–December 2012**.
  - CDS trading volume: **August 2010–December 2012**.

2.2 Net versus gross notional: what the paper is actually measuring

- **Gross notional** adds all protection bought and sold, even if positions offset each other in a chain of counterparties.
- **Net notional** collapses offsetting positions and captures the maximum amount that would need to be transferred in the event of a credit event.
- **Why net notional matters.** It is much closer to the economically meaningful quantity of credit risk transferred.
- **Illustration from Figure 1.** If Y buys \$10m of protection from X, then sells \$10m to Z, and Z sells \$10m to X, gross notional becomes \$30m while net notional becomes **\$0**. This is the paper's way of emphasizing that gross notional can be misleading for economic interpretation.

2.3 Summary statistics that matter

- The paper has **22,229** firm-month observations in the baseline sample for many firm characteristics, and **13,411** firm-month observations with DTCC net/gross notional position data.
- Mean net notional CDS is about **\$1.035 billion**; mean gross notional CDS is about **\$13.13 billion**. Median values are about **\$817 million** and **\$9.914 billion**, respectively.
- Median firm assets are about **\$10.63 billion**; median outstanding bonds are about **\$1.798 billion**.
- Median net CDS as a fraction of outstanding bonds is about **26.1%** using directly issued bonds and **18.2%** when bonds issued by subsidiaries are included.
- Mean monthly CDS turnover is about **50.6%** (median **43.7%**), much higher than mean bond turnover of about **7.16%** (median **4.62%**).
- **Interpretation.** CDS activity is big, but not usually bigger than the underlying stock of insurable interest once one looks at *net* rather than gross positions. And the turnover comparison already hints that CDS is the higher-liquidity venue.

2.4 Key explanatory variables

- **Insurable interest / hedging proxies.**
  - Outstanding bonds.
  - Bonds issued by subsidiaries.
  - Other borrowing (nonbond debt).
  - Accounts payable.
  - Credit enhancement dummy (monoline-style counterparty-risk providers).
- **Speculation proxy.**

$$\text{Disagreement}_{i,t} = \frac{\text{SD of analyst EPS forecasts}_{i,t}}{\text{Stock price}_{i,t}}. \tag{1}$$

  This scales disagreement by the equity cushion and is meant to proxy for disagreement about future default prospects.
- **Standardization / bond-friction proxies.**
  - **Bond fragmentation:** the negative of the log Herfindahl of a firm's bond issues, orthogonalized with respect to total outstanding bonds.
  - **High fragmentation dummy:** indicator for above-median fragmentation.
  - **Contractual heterogeneity dummy:** equals one if bond issues of the same firm differ along features like callability, putability, convertibility, fixed coupon, covenants, or credit enhancement.
- **Liquidity mechanism variables.**
  - **Implied bond roundtrip cost,** from paired TRACE trades.

- **Bond turnover,** monthly bond volume divided by outstanding bonds.
- **Amihud price impact,** an illiquidity measure based on absolute price changes divided by trading volume.

- **Arbitrage variables.**
  - **CDS–bond basis:** CDS spread minus bond spread over the risk-free rate.
  - **Negative basis:** absolute value of the most negative implementable basis for a firm in a month.
  - **Positive basis:** largest implementable positive basis.
  - **Strength of the basis trade:** a time-varying coefficient measuring how sensitive CDS positions are to the negative basis each month.

2.5 Why the sample is econometrically unusual

- DTCC discloses positions only for the **top 1,000** reference entities by notional. That means some firms are known to have CDS markets (because they have Markit quotes), but their net notional positions are not directly observed in DTCC.
- This generates a **censoring problem**, not simple missingness. Figure 4 visualizes this: for some firms, we know net CDS is positive but below the DTCC reporting cutoff.
- As a result, ordinary least squares on observed DTCC positions would be biased. This is why the baseline position regressions are estimated with a censored maximum-likelihood specification.

3 Models and empirical specifications (all equations + interpretation)

3.1 The paper's empirical-hypothesis framework

The paper is not a structural equilibrium estimation of the whole CDS market. Instead it organizes the evidence around four empirical hypotheses:

- **Hedging hypothesis.** More insurable interest should imply larger net CDS positions and more CDS trading volume.
- **Speculation hypothesis.** More disagreement should imply larger net CDS positions and especially larger CDS trading.
- **Standardization hypothesis.** When the underlying bond market is more fragmented and heterogeneous, investors should shift credit-risk trading into CDS.
- **Arbitrage hypothesis.** Deviations from bond–CDS no-arbitrage should be associated with larger CDS positions, especially when arbitrage capital can respond.

3.2 Baseline regression for net CDS positions

The core specification for net notional CDS positions is

$$\text{NetCDS}_{i,t} = \beta X_{i,t} + \varepsilon_{i,t}, \tag{1}$$

where $X_{i,t}$ contains the hedging proxies, disagreement, fragmentation / heterogeneity measures, controls, and a constant.

Interpretation.

- The left-hand side is the *stock* of net credit risk transferred for firm i in month t.
- Because the top-1,000 DTCC disclosure rule censors the dependent variable, the paper estimates this equation by maximum likelihood under censoring.
- The error variance is allowed to scale with outstanding bonds, which is useful because very large firms naturally have more dispersion in notional positions.

3.3 Why this is not a scaled regression

A subtle point in the paper is that it *does not* divide net CDS by assets or by bonds on the left-hand side.

- The authors argue that neither hedging motives, speculation, nor basis-trade demand naturally scale one-for-one with assets.
- Instead they keep the dependent variable in levels and put the relevant scale controls — especially bonds — on the right-hand side.
- This matters because the paper wants to interpret coefficients as marginal dollars of CDS activity associated with marginal dollars of bonds or with a change in fragmentation or disagreement.

3.4 Joint regression for CDS and bond trading volume

To compare the *flow* of trading in CDS and bonds, the paper estimates a joint tobit system:

$$\begin{aligned} \text{CDSVolume}^*_{i,t} &= \beta^{CDS} X_{i,t} + u^{CDS}_{i,t}, \\ \text{BondVolume}^*_{i,t} &= \beta^{Bond} X_{i,t} + u^{Bond}_{i,t}, \end{aligned} \tag{2}$$

with observed volumes censored at zero:

$$\text{Volume}_{i,t} = \max\{0, \text{Volume}^*_{i,t}\}. \tag{3}$$

The two error terms are allowed to be correlated.

Interpretation.

- Many firm-months have zero trading in one or both markets, so the paper treats this as a corner-solution problem rather than discarding zeroes.
- Estimating the bond and CDS equations jointly lets the authors ask directly whether the same shock moves trading in both markets.
- Most importantly, it lets the paper compare the *same right-hand-side variable* across the two markets and test whether its coefficients differ.

3.5 Bond fragmentation as a measure of trading frictions

The bond-fragmentation measure is built from the Herfindahl of bond issues:

$$H_i = \sum_{j \in \text{issues of } i} s_{ij}^2, \tag{4}$$

where s_{ij} is the share of issue j in total outstanding bonds of firm i.

The paper then sets

$$\text{Fragmentation}_i = -\log(H_i) \text{ orthogonalized with respect to } \log(\text{Bonds}_i). \tag{5}$$

Interpretation.

- A firm with many similarly sized bond issues has a low Herfindahl and therefore high fragmentation.
- Orthogonalizing with respect to outstanding bonds removes the mechanical tendency of firms with more debt to have more issues.
- The measure is appealing because it captures a first-order difference between the cash bond market and CDS market: bonds are fragmented and contractually heterogeneous, while CDS contracts are standardized.

3.6 CDS–bond arbitrage specification

The paper defines the CDS–bond basis for a bond as

$$\text{Basis} = \text{CDS spread} - \text{bond spread over the risk-free rate}. \tag{6}$$

A negative basis means the bond is cheap relative to synthetic credit exposure created using a bond plus CDS hedge. The paper then estimates net CDS positions on the negative basis and its interactions:

$$\text{NetCDS}_{i,t} = \beta_1 \, \text{NegBasis}_{i,t} + \beta_2 \, (\text{NegBasis}_{i,t} \times Z_{i,t}) + \beta_3 \, \text{PosBasis}_{i,t} + \Gamma X_{i,t} + \varepsilon_{i,t}, \tag{7}$$

where $Z_{i,t}$ can be the TED spread, bond fragmentation, roundtrip cost, or disagreement.

Interpretation.

- If arbitrageurs buy the bond and buy CDS protection when the basis is negative, then more negative basis should imply larger net CDS.
- The interaction terms test economic mechanisms:
  - NegBasis $\times$ TED: funding stress should weaken basis arbitrage.
  - NegBasis $\times$ Fragmentation or $\times$ RoundtripCost: illiquid bonds make arbitrage more expensive.
  - NegBasis $\times$ Disagreement: directional speculation competes with arbitrage.

3.7 Second-stage measure: strength of the basis trade

To study the economic consequences of arbitrage, the paper constructs a time-varying series

$$\text{StrengthBasisTrade}_t = \hat{\beta}_t^{NB}, \tag{8}$$

where $\hat{\beta}_t^{NB}$ is the month-t cross-sectional coefficient from the regression of net CDS on negative basis (holding the control coefficients common across months).

It then runs second-stage regressions such as

$$\text{NegativeBasis}_{i,t} = a + b\,\text{StrengthBasisTrade}_t + cX_{i,t} + \eta_{i,t}, \tag{9}$$

and

$$\text{AmihudPriceImpact}_{i,t} = \alpha + \delta\,\text{StrengthBasisTrade}_t + \theta X_{i,t} + \nu_{i,t}. \tag{10}$$

Interpretation.

- If basis traders matter, a higher strength-of-basis-trade coefficient should mean the negative basis is less negative and bond price impact is lower.
- Because the strength measure is itself estimated, these second-stage tables use bootstrapped standard errors.

3.8 Liquidity mechanism specification

To show that fragmentation works through bond-market illiquidity, the paper regresses bond trading-cost and turnover measures on fragmentation and contractual heterogeneity:

$$\text{RoundtripCost}_{i,t} = \alpha + \beta_1\text{Fragmentation}_{i,t} + \beta_2\text{Heterogeneity}_{i,t} + \Gamma X_{i,t} + \varepsilon_{i,t}, \tag{11}$$

$$\text{BondTurnover}_{i,t} = \alpha + \beta_1\text{Fragmentation}_{i,t} + \beta_2\text{Heterogeneity}_{i,t} + \Gamma X_{i,t} + \varepsilon_{i,t}. \tag{12}$$

Interpretation.

- This is the *mechanism test*. It says: if CDSs are especially attractive when bonds are fragmented, is that because these bonds are actually more illiquid? The answer is yes.

4 Identification and instruments (what makes the estimates credible)

4.1 What identifies the hedging effect?

- The cleanest hedging proxy is outstanding bonds. CDSs directly reference bonds, so more bonds mechanically create more insurable interest.
- The paper shows that the relation is not just size: when assets are added, assets are small and insignificant, while bonds remain strong.
- It also shows that bonds matter more than other forms of debt such as nonbond borrowing or debt issued by subsidiaries.

4.2 What identifies the speculation effect?

- The main speculation proxy is analyst disagreement normalized by stock price.
- The paper's strongest identifying observation is not just that disagreement predicts CDS activity, but that it predicts *CDS trading much more strongly than bond trading*. That comparative pattern is hard to rationalize with a pure hedging story.

4.3 What identifies the standardization / liquidity effect?

- The key empirical objects are bond fragmentation and contractual heterogeneity.
- Fragmentation is attractive because it is plausibly more exogenous than direct measures such as volume or trading cost. Managers do choose their issuance structure, but they are unlikely to do so in order to manipulate CDS trading on their bonds.
- The paper goes further and uses *predetermined bond fragmentation*, measured when the firm first becomes a traded CDS reference entity (or in early sample), to reduce reverse-causality concerns.
- The mechanism is validated by Table 4: the same firms with fragmented and heterogeneous bonds are the ones whose bonds have higher trading costs and lower turnover.

4.4 What identifies the arbitrage channel?

- The negative basis creates a sharp cross-sectional prediction: when bond prices are cheap relative to CDS, arbitrageurs should buy bonds and buy CDS protection, raising net CDS.
- The interaction with the TED spread is particularly convincing. If the relation between negative basis and net CDS reflects arbitrage, it should weaken when funding conditions are tight. This is exactly what the paper finds.
- The second-stage "strength of the basis trade" series then links quantities to prices and liquidity outcomes: when basis positions respond more strongly to the negative basis, the basis is smaller and bond price impact is lower.

4.5 Why the estimates are descriptive, not clean causal

- The authors are explicit that the paper is *largely descriptive*. There are no natural experiments that sharply isolate one motive from another.
- Some right-hand-side variables are still endogenous. For example, disagreement, bond trading costs, and the basis are all market outcomes as well as explanatory variables.
- The paper addresses this by emphasizing comparative predictions and mechanisms rather than claiming a single clean causal estimate.

4.6 Main limitations

- The sample is restricted to public U.S. parent companies with ratings, analyst coverage, and bond issuance, so it is not the full universe of CDS reference entities.

- Net notional data are censored because DTCC only reports the top 1,000 names.
- Contractual heterogeneity is measured coarsely using bond-feature dummies from FISD.
- The basis analysis uses only firms for which an implementable basis can be computed, which is a much smaller subset.
- The results are strongest in the post-crisis 2008–2012 environment; that is the best possible period for DTCC data, but also a period in which the basis trade and liquidity conditions were unusually salient.

4.7 Relation to the literature

- The paper connects to the theory literature on the uses of CDSs: hedging, speculation, loan sales versus CDS, empty creditors, and equilibrium effects on bond markets.
- It also connects to the empirical literature on CDS effects on information transmission, risk transfer, and corporate financing.
- Its sharpest contribution is to the CDS–bond basis literature: rather than focusing only on prices, it links *quantities* in the CDS market to basis arbitrage and then links this arbitrage to price impact in the bond market.

5 Tables and figures

5.1 Figure 1 and Table 1: what the data actually measure

Read:

- Figure 1 explains the difference between gross and net notional CDS with a simple three-counterparty example.
- Table 1 gives the central magnitudes: mean net CDS of about **$1.035bn**, mean gross CDS of about **$13.13bn**, median net CDS of about **$817m**, and median gross CDS of about **$9.914bn**.
- Median net CDS equals about **26.1%** of directly issued bonds and about **18.2%** if bonds issued by subsidiaries are included.
- Mean monthly CDS turnover is about **50.6%**, versus bond turnover of about **7.16%**.

Why it matters.

- Figure 1 tells you why the paper focuses on net rather than gross notional.
- Table 1 shows CDS markets are large but that net positions usually do *not* vastly exceed the firm's insurable interest.
- The turnover numbers foreshadow the paper's liquidity/standardization interpretation.

5.2 Figure 2 and Figure 3: market size and trading activity

Read:

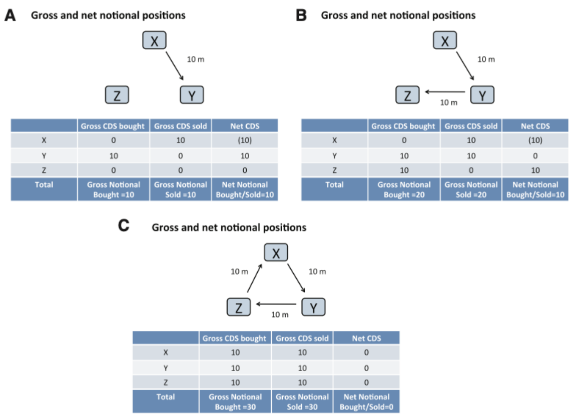

Figure 1
Gross notional and net notional CDS positions
This figure illustrates the difference between gross notional and net notional amounts in the DTCC data. Y has purchased $10 m in protection from X (*A*). Both the gross notional and the net notional amount outstanding are $10 m. Y offsets the initial trade by selling $10 m in protection to Z (*B*). This raises the gross notional amount to $20 m. The net notional amount remains at $10 m. Z sells $10 m in protection to X, such that all three parties have a net zero position (*C*). The gross notional is now $30 m but, because all net positions are zero, the net notional is $0.

Table 1
Summary statistics

	(1)	(2)	(3)	(4)	(5)	(6)
VARIABLES	N	Mean	SD	p10	p50	p90
Assets	22,229	51.22	191.3	2.641	10.63	79.55
Debt	22,229	14.13	65.82	0.690	3.005	17.08
Bonds	22,229	4.505	12.66	0.400	1.798	9.252
Bonds issued by subsidiaries	22,229	2.452	9.046	0	0	4.908
Other borrowing	22,229	7.861	50.63	0	0.321	5.670
Accounts payable	22,229	1.462	3.876	0	0.355	3.379
Credit enhancement (dummy)	22,229	0.00787	0.0884	0	0	0
Net CDS	13,411	1.035	0.869	0.317	0.817	1.870
Gross CDS	13,411	13.13	12.19	2.890	9.914	25.40
Net CDS / bonds (direct issue)	13,411	0.503	0.713	0.0758	0.261	1.138
Net CDS / bonds (incl. subsidiary)	13,411	0.364	0.535	0.0505	0.182	0.859
5y CDS spread (bps)	19,705	249.2	369.0	50.71	138.8	541.3
Negative basis (%)	3,012	0.917	1.496	0	0.496	2.206
Positive basis (%)	3,012	0.206	0.575	0	0	0.554
CDS volume (monthly)	7,781	0.516	0.646	0.0594	0.334	1.097
CDS turnover (monthly)	7,781	0.506	0.344	0.142	0.437	0.950
Bond volume (monthly)	22,229	0.370	1.094	0.00261	0.0916	0.766
Bond turnover (monthly)	22,229	0.0716	0.0873	0.00456	0.0462	0.159
Implied bond roundtrip cost (%)	15,405	0.369	0.323	0.105	0.280	0.723
Issuer bond Herfindahl	22,229	0.330	0.271	0.0903	0.240	1
Bond fragmentation	22,229	0.0191	0.406	−0.525	0.0447	0.495
S&P rating (notch)	22,229	9.212	3.042	6	9	13
Disagreement: analyst SD/price	22,229	0.0135	0.0313	0.00147	0.00552	0.0260
Contractual heterogeneity (dummy)	22,229	0.671	0.470	0	1	1
Amihud price impact	15,639	3.132	4.056	0.677	2.023	6.522

This table provides the summary statistics for our data: the number of observations, mean, standard deviation, and 10th, 50th, and 90th percentile. Our sample consists of U.S. public parent companies in Compustat that are traded reference entities in the CDS market, have at least one outstanding bond, are rated by S&P, and for which the *Disagreement: analyst SD/price* measure can be calculated. Data are monthly from October 2008 to December 2012, with CDS trading data starting in August 2010. We categorize a firm as a traded reference entity in the CDS market once the firm appears in the DTCC data or, in a given month, has at least three quotes from dealers in the Markit data. The variables are defined as follows. *Assets* is total assets from Compustat. *Debt* is the sum of long-term and short-term debt from Compustat. *Bonds* is the total face value of bonds issued directly by the parent company with at least one year of maturity at issuance from Mergent FISD. *Bonds issued by subsidiaries* is the total face value of bonds issued by subsidiaries of the parent with at least one year of maturity at issuance from Mergent FISD. *Other borrowing* is nonbond debt defined as *debt*, minus the sum of *bonds* and *bonds issued by subsidiaries*. *Accounts payable* is from Compustat. We set accounts payable to zero for financial firms, because it includes deposits and is thus not comparable to nonfinancials. *Credit enhancement (dummy)* is a dummy variable indicating firms providing credit enhancement. *Net CDS* and *gross CDS* are the net and gross notional CDS amounts, respectively, as reported by the DTCC. *5y CDS spread (bps)* is the five-year CDS spread from Markit in basis points. *Negative basis* and *positive basis* are the absolute value of the smallest daily negative CDS-bond basis and the largest daily positive CDS-bond basis across issuances of the same firm calculated using implementable trades of at least $1 million in face value, averaged for the month. Firm-months for which a basis can be calculated but without a positive maximum or negative minimum are set to zero. *CDS volume (monthly)* is total monthly CDS trading volume that constitutes market risk activity, in billions of dollars of notional as reported by the DTCC. *CDS turnover (monthly)* is *CDS volume*, divided by *net CDS*. *Bond volume (monthly)* is the total monthly bond trading volume in billions of dollars of notional for all bonds a company has issued, taken from Trace Enhanced. *Bond turnover (monthly)* is *bond volume*, divided by *bonds*. *Implied bond roundtrip cost* is the implied roundtrip cost of paired bond trades from Trace Enhanced with trade volume above $1 million. *Issuer bond Herfindahl* is the Herfindahl measure of a firm's bond issues based on Mergent FISD. *Bond fragmentation* is negative *log(issuer bond Herfindahl)* orthogonalized with respect to *log(bonds)*. *S&P rating (notch)* captures a firm's S&P rating; it takes a value of 1 for AAA, 2 for AA+, etc. The maximum value, 22, indicates that the bond has defaulted. *Disagreement: analyst SD/price* is defined as the standard deviation of analyst earnings forecasts from IBES, normalized by the stock price from CRSP. *Contractual heterogeneity (dummy)* is an indicator variable of whether all bonds of an issuer have the same contractual features along six dimensions: callable, puttable, convertible, fixed coupon, covenants, and credit enhancement. *Amihud price impact* is the absolute value of the percentage change in the mid-price of a firm's specific bond issue from the previous day $t-1$ to the next trading day $t+1$, divided by bond trading volume on that trading day t (in $ billions if volume exceeds $1 million), averaged across all days and bond issues of a given firm in a given month. All dollar amounts are in billions.

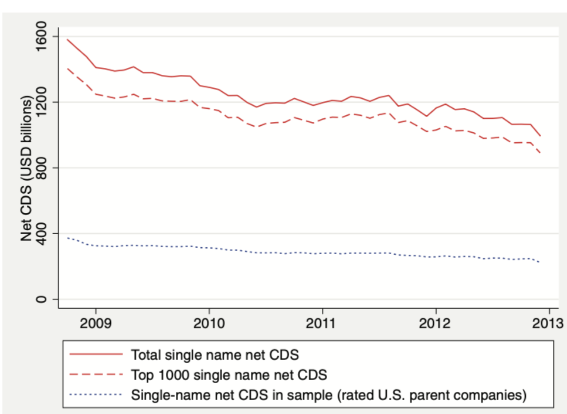

Figure 2
Single-name net notional CDS amounts over time
This figure plots single-name net notional CDS amounts over time. The top solid line plots the total net notional CDS amount on all single-name reference entities, as reported by the DTCC. It thus captures the entire single-name CDS market. The dashed line plots the total net notional amount of CDS protection written on the top 1,000 single-name reference entities. Comparing this line with the total single-name CDS market demonstrates that the top 1,000 reference entities make up a large fraction of the single-name CDS market when measured in terms of net notional. The dotted line plots the total net notional CDS amount for reference entities in our sample: rated public U.S. parent companies in Compustat with at least one outstanding bond.

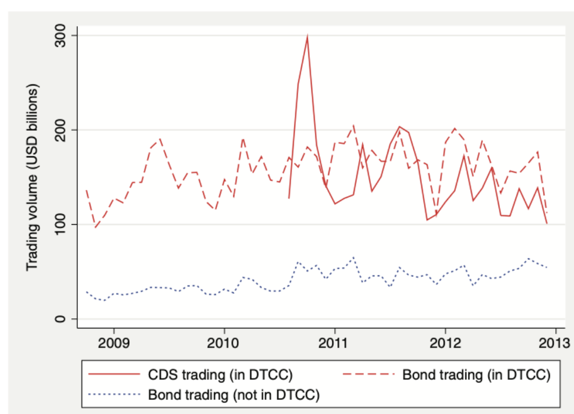

Figure 3
Trading volume in CDSs and bonds
This figure plots trading volume in the bond market and the CDS market. The solid line depicts monthly trading volume in the CDS market as measured by trades that represent market risk activity according to the DTCC. The dashed line depicts monthly trading volume of directly issued bonds for firms for which we also have CDS trading volume (DTCC companies). The dotted line represents trading volume of directly issued bonds for non-DTCC companies.

- Figure 2 shows total single-name net CDS declining by about **35%** from late 2008 to end-2012, but still around **$1 trillion** at the end of the sample.
- The sample of matched rated U.S. public parent companies remains a substantial fraction of the corporate single-name market.
- Figure 3 compares monthly CDS trading volume and bond trading volume. For traded reference entities, the levels are broadly comparable, but CDS turnover is much higher because net CDS outstanding is smaller than the stock of bonds.

Why it matters.

- These figures show the paper studies an economically important market, not a niche corner.
- They also visually motivate the idea that CDS is a fast-turnover venue for credit-risk trading.

5.3 Figure 4: the censoring problem

Read:

- Figure 4 plots net CDS against log outstanding bonds for a sample month.
- Some firms are observed in DTCC; others have evidence of a CDS market from Markit quotes but are not in DTCC's top 1,000. For these firms, we know CDS exists but net notional is below the disclosure threshold.

Why it matters.

- This figure justifies the censored-regression framework. The paper does not face ordinary missing data; it faces a top-coded reporting design.

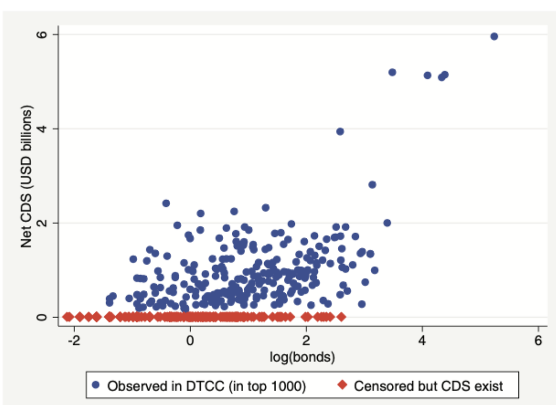

Figure 4
Censoring in net notional CDS amounts
This figure illustrates the censoring in our sample, based on one example month, December 2009. We plot *net CDS* as a function of *log(bonds)* for all companies for which we know that the CDS market exists: companies that we observe in the DTCC data and/or for which we can find a Markit CDS quote from at least three dealers. Circles denote firms in the DTCC data, and diamonds denote firms not in DTCC. For expositional purposes, for firms not in DTCC, we set the net notional CDS amount to zero in the figure. However, we know that these firms have positive net notional CDS amounts (but do not make it into the top 1,000 reference entities).

Table 2
Net notional CDS amounts

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
	Net CDS	Net CDS	Net CDS	Net CDS	Net CDS	Net CDS	Net CDS	Net CDS	Net CDS
Bonds	0.0777***	0.0138	0.0651***	0.0874***	0.0772***	0.0764***	0.0875***	0.0692***	0.0705***
	(7.30)	(1.36)	(5.34)	(7.15)	(7.43)	(7.40)	(6.91)	(7.13)	(7.16)
Disagreement: analyst SD/price	4.672***	0.928**	4.645***	2.697***	4.622***	4.595***	4.945***	4.789***	4.734***
	(4.74)	(2.31)	(4.68)	(3.00)	(4.77)	(4.70)	(4.32)	(4.82)	(4.83)
Bond fragmentation					0.272***				0.181**
					(3.41)				(2.16)
High fragmentation (dummy)						0.272***			
						(4.27)			
Predetermined bond fragmentation							0.165**		
							(2.29)		
Contractual heterogeneity (dummy)								0.291***	0.235***
								(3.97)	(3.04)
Assets			0.000936						
			(1.21)						
Bonds issued by subsidiaries				0.00732					
				(1.10)					
Other borrowing				−0.00535**					
				(−2.06)					
Accounts payable				0.0163**					
				(2.52)					
Credit enhancement (dummy)				2.215***					
				(4.63)					
Rating controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Time fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Industry fixed effects	Yes	No	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Firm fixed effects	No	Yes	No	No	No	No	No	No	No
Number of firms	496	302	496	496	496	496	431	496	496
Number of observations	22,229	13,946	22,229	22,229	22,229	22,229	19,606	22,229	22,229

This table presents the results of a censored regression of *net CDS* on proxies for hedging, speculation, bond market liquidity, and controls. For variable definitions, see Table 1. *High fragmentation (dummy)* is an indicator variable if bond fragmentation is above the median. *Predetermined bond fragmentation* is defined as *Bond fragmentation* in the month that the firm becomes a traded reference entity in the CDS market or in January 2002 (whichever is later). All regressions include time and rating dummies. The *Rating controls* are dummies for AA and above, A, BBB, B, CCC, and below. BB is the omitted control. Column (2) includes firm fixed effects, all other columns include industry dummies based on first-digit SIC codes. Note that the constant is subsumed by the control dummies. When including firm fixed effects, we only include companies that appear in the DTCC data at least once. The sample period is October 2008 to December 2012. We jointly estimate heteroscedasticity of the error term in the functional form of constant, plus a coefficient times *bonds*. *t*-statistics are given in parentheses based on standard errors clustered by firm. ***, **, and * denote significance at the 1%, 5%, and 10% level, respectively.

5.4 Table 2: determinants of net CDS positions

Read:

- In the baseline specification, each extra dollar of outstanding bonds is associated with about **7.77 cents** more net CDS.
- Moving from the 25th to the 75th percentile of outstanding bonds raises net CDS by about **$261m**, which is about **31.9%** of the median firm's net CDS.
- Analyst disagreement significantly raises net CDS; one standard deviation more disagreement adds about **$146m**, roughly **18%** of the median firm's net CDS.
- Credit-enhancement firms carry about **$2bn** more net CDS, consistent with hedging of counterparty risk.
- Accounts payable also matters positively, suggesting trade-credit exposure motivates hedging.
- Above-median bond fragmentation is associated with about **$272m** more net CDS.
- Contractual heterogeneity is associated with about **$291m** more net CDS.

Why it matters.

- This table establishes the paper's three main motives in one place: hedging, speculation, and standardization.
- The bond-fragmentation and contractual-heterogeneity results are the cleanest evidence that CDS markets substitute for a less standardized cash bond market.

5.5 Table 3: CDS trading volume versus bond trading volume

Read:

- Outstanding bonds raise trading volume in both markets:
  - about **8.62 cents** more monthly *bond* trading per extra dollar of bonds,
  - about **2.83 cents** more monthly *CDS* trading per extra dollar of bonds.
- Disagreement has a large effect in CDS but not in bonds. One standard deviation more disagreement raises monthly CDS trading by about **$144m**, roughly **43.1%** of median CDS trading volume, but has only a small and statistically insignificant effect on bond trading.
- Above-median fragmentation means about **$89.9m** more CDS trading but about **$20.1m** less bond trading.
- Contractual heterogeneity means about **$84m** more CDS trading but about **$17.3m** less bond trading.

Why it matters.

- Hedging-related activity shows up in both markets.
- Speculation is disproportionately concentrated in the more liquid CDS market.
- Fragmentation shifts the *location* of credit-risk trading away from bonds and toward CDS. This is one of the central empirical claims of the paper.

Table 3
Bond and CDS trading volume

	(1)		(2)		(3)		(4)		(5)	
	CDS volume	Bond volume	CDS volume	Bond volume	CDS volume	Bond volume	CDS volume	Bond volume	CDS volume	Bond volume
Bonds	0.0283***	0.0862***	0.0156***	0.164***	0.0284***	0.0896***	0.0281***	0.0881***	0.0269***	0.0893***
	(8.94)	(29.72)	(2.79)	(11.90)	(9.06)	(28.54)	(9.01)	(29.61)	(8.52)	(29.38)
Disagreement: analyst SD/price	4.594***	0.427	3.352***	0.208	4.831***	0.350	4.890***	0.391	4.825***	0.367
	(3.29)	(1.14)	(2.94)	(1.08)	(3.42)	(0.90)	(3.44)	(1.08)	(3.55)	(1.04)
Bond fragmentation					0.0825**	−0.0377***				
					(2.05)	(−3.88)				
High fragmentation (dummy)							0.0899***	−0.0201***		
							(2.74)	(−3.45)		
Contractual heterogeneity (dummy)									0.0843**	−0.0173***
									(2.52)	(−2.97)
Rating controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Time fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Industry fixed effects	Yes	Yes	No	No	Yes	Yes	Yes	Yes	Yes	Yes
Firm fixed effects	No	No	Yes	Yes	No	No	No	No	No	No
Number of firms	288		287		288		288		288	
Number of observations	8,041		8,029		8,041		8,041		8,041	
p-value: coefficients on bonds equal	0.000		0.000		0.000		0.000		0.000	
p-value: coefficients on disagreement equal	0.005		0.007		0.002		0.003		0.002	
p-value: coefficients on bond fragmentation equal					0.004		0.001			
p-value: coefficients on contr. heterogeneity equal									0.002	

Each pair of columns reports the results of a joint tobit regression of *CDS volume* (left column) and *bond volume* (right column) on proxies for hedging, speculation, bond market liquidity, and controls. For variable definitions, see Table 1, and for control definitions see, Table 2. The sample consists of firms with net CDS reported in the DTCC data. Note that the constant is subsumed by the control dummies. The sample period is August 2010 to December 2012. We jointly estimate heteroscedasticity of the error term in both regressions in the functional form of constant, plus a coefficient times *bonds*. The error terms in the two regressions are allowed to be correlated. t-statistics are given in parentheses based on standard errors clustered by firm. ***, **, and * denote significance at the 1%, 5%, and 10% level, respectively. At the bottom of the table we present p-values for the hypothesis that coefficients are equal across the CDS and the bond market.

Table 4
Bond fragmentation and bond liquidity

	(1)	(2)	(3)	(4)	(5)	(6)
	Roundtrip cost	Roundtrip cost	Roundtrip cost	Turnover	Turnover	Turnover
Bond fragmentation	0.0426***		0.0388***	−0.0199***		−0.0186***
	(3.90)		(3.45)	(−3.98)		(−3.85)
Contractual heterogeneity (dummy)		0.0246**	0.0121		−0.0101**	−0.00398
		(2.46)	(1.17)		(−2.54)	(−1.13)
log(bonds)	−0.00881*	−0.0124**	−0.0108**	0.0154***	0.0171***	0.0162***
	(−1.70)	(−2.25)	(−1.99)	(8.98)	(9.60)	(9.07)
Disagreement: analyst SD/price	2.145***	2.146***	2.146***	0.0948**	0.0911**	0.0933**
	(7.26)	(7.41)	(7.28)	(2.14)	(1.99)	(2.12)
Time fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Industry fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Rating controls	Yes	Yes	Yes	Yes	Yes	Yes
Number of firms	487	487	487	496	496	496
Number of observations	15,405	15,405	15,405	22,229	22,229	22,229
R-squared	0.240	0.238	0.240			

Columns (1)-(3) present the results of OLS regressions of ***Implied bond rountrip cost*** on measures of bond market fragmentation, contractual heterogeneity, and controls. Columns (4)-(6) present the results of tobit regressions of ***Bond turnover*** on measures of bond market fragmentation, contractual heterogeneity, and controls. For variable definitions, see Table 1 and for control definitions, see Table 2. Note that the constant is subsumed by the control dummies. The sample period is October 2008 to December 2012. t-statistics are given in parentheses based on heteroscedasticity-consistent standard errors clustered by firms. ***, **, and * denote significance at the 1%, 5%, and 10% level, respectively.

5.6 Table 4: does fragmentation actually mean illiquidity?

Read:

- A one-standard-deviation increase in bond fragmentation raises roundtrip trading costs by about **6.2%** relative to the median firm and lowers bond turnover by about **17.5%**.
- Contractual heterogeneity raises roundtrip costs by about **8.8%** and lowers turnover by about **21.9%** relative to the median firm.

Why it matters.

- This is the mechanism table. It says the same bond structures that predict more CDS activity are also the ones that make the bond market less liquid.
- So the "CDS as alternative venue" interpretation is not just rhetoric; it is anchored in actual cash-market illiquidity.

5.7 Table 5 and Figure 5: arbitrage through the basis trade

Read:

- The coefficient on the negative basis is positive and significant across specifications. In the baseline, it is about **0.0888**.
- A one-standard-deviation increase in the negative basis (about **149.6 bps**) is associated with about **$133m** more net CDS, or roughly **16%** of the median firm's net CDS.
- Positive basis is generally insignificant.
- The interaction of negative basis with the TED spread is negative, meaning arbitrage weakens when funding conditions tighten.
- The interaction with roundtrip costs is negative, and the interaction with disagreement is also negative, consistent with illiquidity and speculation crowding out basis arbitrage.
- Figure 5 plots the *strength of the basis trade*. It is near zero in late 2008 / early 2009, rises as funding conditions normalize, and has mean about **0.265** and standard deviation about **0.198**.

Table 5
Net notional CDS amounts and the CDS-bond basis

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	Net CDS	Net CDS	Net CDS	Net CDS	Net CDS	Net CDS	Net CDS
Negative basis	0.0888***	0.0290***	0.203***	0.0923***	0.179***	0.127***	0.267***
	(2.61)	(2.86)	(4.16)	(2.88)	(3.39)	(3.11)	(4.40)
TED spread * neg. basis			−0.104***				−0.0840***
			(−3.87)				(−3.73)
Fragmentation * neg. basis				−0.104			−0.158**
				(−1.34)			(−2.08)
Roundtrip cost * neg. basis					−0.123**		−0.0729**
					(−2.48)		(−2.12)
Disagreement * neg. basis						−1.029***	−0.721
						(−2.90)	(−1.29)
Positive basis	−0.0291	0.00230	0.0523	−0.0700	0.0280	−0.000358	−0.0495
	(−0.23)	(0.09)	(0.39)	(−0.49)	(0.13)	(−0.00)	(−0.18)
TED spread * pos. basis			−0.0926				0.103
			(−1.12)				(1.20)
Fragmentation * pos. basis				0.490*			0.420
				(1.70)			(1.23)
Roundtrip cost * pos. basis					−0.0794		−0.0473
					(−0.88)		(−0.54)
Disagreement * pos. basis						−0.238	−0.270
						(−0.34)	(−0.36)
Bond fragmentation				0.258			0.275
				(1.35)			(1.42)
Implied bond roundtrip cost					0.329**		0.204*
					(2.47)		(1.67)
Disagreement: analyst SD/price	7.656***	2.354***	7.767***	7.054***	7.902***	9.047***	8.825***
	(4.95)	(4.66)	(5.29)	(4.85)	(3.15)	(7.13)	(2.63)
Bonds	0.0524***	0.0000943	0.0523***	0.0542***	0.0516***	0.0527***	0.0525***
	(5.14)	(0.01)	(5.14)	(5.15)	(4.75)	(5.11)	(4.72)
Rating controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Time fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Industry fixed effects	Yes	No	Yes	Yes	Yes	Yes	Yes
Firm fixed effects	No	Yes	No	No	No	No	No
Number of firms	123	97	123	123	120	123	120
Number of observations	2,730	2,422	2,730	2,730	2,423	2,730	2,423

This table presents the results of a censored regression of *Net CDS* on the *Positive basis* and *Negative basis* and proxies for hedging, speculation, bond liquidity, and controls. For variable definitions, see Table 1 and for control definitions, see Table 2. We calculate the basis for firms with CDS quotes from at least three dealers in Markit and at least one fixed coupon bond without embedded options that matures within 1-30 years. We calculate the basis using an implementable basis trade: First, we use only bond trades of at least $1 million in face value. Second, we calculate the positive basis based on the bond bid price and and the negative basis based on the bond ask price. Third, for the negative basis, we take the minimum among bonds

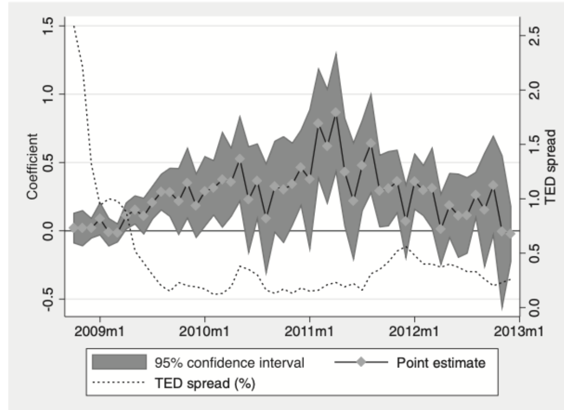

Figure 5
The strength of the basis trade
This figure plots the *Strength of the basis trade*, which is defined as the time-series of the cross-sectional regression coefficient of net notional CDS amounts on the *Negative basis* (left axis). The regression specification is the same as Column (1) of Table 5 but allows for time variation in the coefficient of both the negative and the positive basis. The gray shaded region denotes the 95% confidence interval. The mean of the *Strength of the basis trade* is 0.265, and the standard deviation is 0.198. The dashed line plots funding conditions as measured by the *TED spread* (right axis).

Why it matters.

- Table 5 is the paper's bridge from "CDSs are a trading venue" to "CDSs are part of a cross-market arbitrage technology."
- Figure 5 then converts this into a time-varying quantity that can be used to study how arbitrage affects prices and liquidity.

5.8 Table 6: arbitrage compresses the negative basis

Read:

- The coefficient on the strength of the basis trade is significantly negative.
- Using Column (2), a one-standard-deviation increase in basis-trade strength reduces the negative basis by about **52 basis points**.
- The effect is especially strong during the crisis period.

Why it matters.

- This is the price implication of the arbitrage channel: when arbitrageurs are more active, mispricing between bonds and CDS is smaller.
- That gives CDS markets a broader financing role because compressing a very negative basis effectively supports bond prices.

5.9 Table 7: basis traders reduce bond-market price impact

Read:

- The coefficient on basis-trade strength is significantly negative in the Amihud price-impact regressions.

Table 6
The negative basis and the strength of the basis trade

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
	Neg. basis	Neg. basis	Neg. basis	Neg. basis	Neg. basis	Neg. basis	Neg. basis	Neg. basis	Neg. basis
Strength of basis trade	−2.895***	−2.624***	−2.210***	−0.282**	−0.291***	−0.242*	−0.273**	−0.279***	−0.229*
	(−4.75)	(−4.91)	(−5.09)	(−2.51)	(−2.74)	(−1.94)	(−2.35)	(−2.58)	(−1.80)
Strength of basis trade * crisis				−7.004***	−7.128***	−6.873***	−6.128***	−5.676***	−5.096***
				(−4.27)	(−4.40)	(−4.60)	(−2.94)	(−2.91)	(−2.74)
TED spread							0.0594	0.0759	0.0871
							(0.25)	(0.33)	(0.38)
TED spread * crisis							0.130	0.238	0.310
							(0.17)	(0.32)	(0.43)
Crisis (dummy)				2.478***	2.398***	2.376***	2.230***	1.979***	1.860***
				(7.00)	(6.91)	(6.76)	(3.24)	(3.08)	(2.88)
log(bonds)		−0.0909	−0.301*		0.0151	0.452**		0.0173	0.461**
		(−1.51)	(−1.68)		(0.34)	(2.32)		(0.39)	(2.29)
Disagreement: analyst SD/price		15.34**	20.43**		7.737	10.87		7.930	11.35*
		(2.19)	(2.39)		(1.45)	(1.61)		(1.48)	(1.68)
Bond fragmentation		0.173			0.220**			0.227	
		(1.56)			(2.26)			(1.43)	
Constant	2.046***	2.374***		0.828***	1.306***		0.809***	1.276***	
	(8.08)	(7.67)		(18.81)	(7.42)		(9.09)	(8.13)	
Industry fixed effects	No	Yes	No	No	Yes	No	No	Yes	No
Rating controls	No	Yes	Yes	No	Yes	Yes	No	Yes	Yes
Firm fixed effects	No	No	Yes	No	No	Yes	No	No	Yes
Number of dates	51	51	51	51	51	51	51	51	51
Number of firms	120	120	120	120	120	120	120	120	120
Number of observations	2,112	2,112	2,112	2,112	2,112	2,112	2,112	2,112	2,112
R-squared	0.116	0.239	0.439	0.303	0.397	0.561	0.304	0.400	0.565

This table presents the results of an OLS regression of the *Negative basis* on the *Strength of the basis trade* and proxies for hedging, speculation, bond liquidity, and controls. Since the *Negative basis* is an estimated variable, this regression is the second step in a two-step estimation. The sample only includes firm-month observations for firms with at least one bond with a negative basis that month. The *Strength of the basis trade* is graphed and described in Figure 5. For variable definitions, see Table 1, and for control definitions, see Table 2, for the definition of the basis see Table 5. The sample period is October 2008 to December 2012. t-statistics, based on bootstrapped standard errors calculated from 1,000 bootstrapped samples drawn by first resampling dates and then resampling firms within each date, are given in parentheses. We then apply the two-step estimation procedure to each bootstrapped sample to calculate the standard deviation of the estimated coefficients. ***, **, and * denote significance at the 1%, 5%, and 10% level, respectively.

Table 7
Price impact in the bond market and the strength of the basis trade

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
	Amihud price impact	Amihud price impact	Amihud price impact	Amihud price impact	Amihud price impact	Amihud price impact	Amihud price impact	Amihud price impact	Amihud price impact
Strength of basis trade	−3.597***	−2.954***	−2.677***	0.126	0.0753	−0.0214	0.374	0.348	0.258
	(−3.70)	(−3.60)	(−3.95)	(0.33)	(0.20)	(−0.06)	(1.04)	(1.02)	(0.77)
Strength of basis trade * crisis				−14.23***	−12.67***	−12.28***	−5.221**	−3.390*	−2.870
				(−4.46)	(−4.11)	(−4.17)	(−2.54)	(−1.87)	(−1.52)
TED spread							1.617**	1.792***	1.850***
							(2.38)	(2.79)	(2.85)
TED spread * crisis							0.426	0.325	0.323
							(0.44)	(0.37)	(0.38)
Crisis (dummy)				4.675***	4.060***	3.842***	2.369***	1.714***	1.462**
				(6.50)	(5.77)	(5.38)	(3.39)	(2.85)	(2.37)
log(bonds)		−0.648***	−1.615***		−0.553***	−0.624***		−0.560***	−0.581**
		(−9.97)	(−6.14)		(−10.63)	(−2.67)		(−10.98)	(−2.33)
Disagreement: analyst SD/price		35.05***	39.56***		25.61***	27.37***		25.43***	27.38***
		(8.07)	(7.32)		(6.56)	(5.32)		(6.72)	(5.73)
Bond fragmentation		0.693***	−0.194		0.735***	0.131		0.751***	0.108
		(5.06)	(−0.36)		(5.83)	(0.26)		(6.77)	(0.22)
Constant	4.112***	3.894***		2.373***	2.639***		1.848***	2.053***	
	(9.91)	(10.91)		(18.25)	(15.34)		(8.36)	(14.26)	
Industry fixed effects	No	Yes	No	No	Yes	No	No	Yes	No
Rating controls	No	Yes	Yes	No	Yes	Yes	No	Yes	Yes
Firm fixed effects	No	No	Yes	No	No	Yes	No	No	Yes
Number of dates	51	51	51	51	51	51	51	51	51
Number of firms	469	469	469	469	469	469	469	469	469
Number of observations	15,639	15,639	15,639	15,639	15,639	15,639	15,639	15,639	15,639
R-squared	0.0305	0.133	0.259	0.126	0.200	0.311	0.142	0.218	0.328

This table presents the results of an OLS regression of price impact in the bond market on the strength of the basis trade. The left-hand-side variable is the *Amihud price impact* measure of illiquidity of the bond market for a given firm in a given month. The right-hand-side variables are the *Strength of the basis trade* and proxies for hedging, speculation, bond liquidity, and controls. Since the *Negative basis* is an estimated variable, this regression is the second step in a two-step estimation. To define *Amihud price impact*, we first calculate the absolute value of the percentage change in the mid-price of a firm's specific bond issue from the previous day $t-1$ to the next trading day $t+1$, divided by bond trading volume on that trading day t (in $ billions if volume exceeds $1 million). We then average this across all days and bond issues of a given firm in a given month. This results in a similar measure of illiquidity as the one introduced by Amihud (2002) for stocks. The *Strength of the basis trade* is graphed and described in Figure 5. For variable definitions, see Table 1, and for control definitions, see Table 2. The sample period is October 2008 to December 2012. t-statistics, based on bootstrapped standard errors calculated from 1,000 bootstrapped samples drawn by first resampling dates and then resampling firms within each date, are given in parentheses. We then apply the two-step estimation procedure to each bootstrapped sample to calculate the standard deviation of the estimated coefficients. ***, **, and * denote significance at the 1%, 5%, and 10% level, respectively.

- Using Column (2), a one-standard-deviation increase in basis-trade strength lowers the Amihud price-impact measure by about **28.9%** for the median firm.
- The effect is concentrated in the crisis period.

Why it matters.

- This is the market-liquidity implication of the paper. Basis traders are not just correcting relative prices; they are also absorbing supply shocks in the bond market.
- So CDS markets can indirectly improve cash-bond market resilience.

6 Short summary

- **Method.** Use entity-level DTCC position and trading data, matched to bond-market structure and liquidity measures, and test hedging, speculation, standardization, and arbitrage hypotheses.
- **Main facts.** More bonds ⇒ more CDS positions. More disagreement ⇒ much more CDS trading but not much more bond trading. More bond fragmentation / heterogeneity ⇒ more CDS activity and less bond trading. More negative basis ⇒ more net CDS.
- **Mechanism.** Fragmented and heterogeneous bonds are illiquid and expensive to trade. CDS contracts are standardized and more liquid, so investors migrate credit-risk trading there.
- **Arbitrage implication.** Basis traders use CDSs to exploit negative basis trades, compress the basis, and reduce bond-market price impact.
- **Bottom line.** CDS markets are best understood as *alternative trading venues for credit risk*, not just insurance instruments.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

- The paper establishes that CDS activity reflects several motives at once, but the most distinctive economic role of the CDS market is to provide a *standardized and liquid venue* for trading credit risk when the cash bond market is fragmented and illiquid.

- It also establishes that CDS quantities contain information about arbitrage activity linking bond and CDS markets, and that this arbitrage activity has real effects on pricing and liquidity in the bond market.

7.2 Economic intuition

- A corporate bond market is messy: issuers split debt into many issues, each with different coupons, callability, covenants, and maturity. Trading a specific bond can therefore be difficult and expensive.
- A CDS contract strips out the credit-risk component into a standardized derivative. That makes it attractive both for hedgers and for speculators with shorter horizons.
- When a bond becomes unusually cheap relative to CDS, arbitrageurs can buy the bond and buy protection, using the CDS market as one leg of the trade. This helps push bond and CDS pricing back together and cushions bond-market supply shocks.

7.3 Takeaway

- The paper's big contribution is interpretive. It changes how one should think about CDS markets.
- The right picture is not "CDS versus bonds" as two redundant markets. It is "CDS as a cleaner, more standardized layer of credit-risk trading" sitting on top of a fragmented OTC bond market.
- That is why hedging and speculation both show up in CDS markets, and why CDSs can also improve bond-market outcomes through basis-trade arbitrage.